



Module Title

Cybersecurity and Networking

Internship Assignment Submission

Linux Files, Directories & Permissions

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1.THEORY

1.1 Answer all 20 questions in your own words

1.What is a Linux directory?

A directory is a folder used to organize and store files and other directories.

2. What does mkdir do?

Mkdir creates a new directory (folder).

3. What does touch do?

Touch creates an empty file if it does not already exist.

4. What does cat do?

Cat displays the contents of a file in the terminal.

5. What does cp do?

Cp copies file or directory to another location or name.

6. What does mv do?

Mv moves a file or directory. It can also be used to rename them.

7. What does rm do?

Rm removes or deletes file.

8. What does `ls -l` display?

`ls -l` shows detailed information about files, such as permissions, owner, group, size, date, and name.

9. Explain r, w and x permissions.

r means read, w means write, and x means execute.

10. Explain user, group and others.

User is the file owner, group means users in file's group, and others means everyone else.

11. What does `chmod` do?

`Chmod` changes the permissions of a file directory.

12. What does `chmod 755` represent?

The owner has read, write and execute permissions. The group and others have read-only permissions.

13. What does `chmod 644` represent?

The owner has read and write permissions. The group and others have read-only permissions.

14. What is file ownership?

File ownership identifies the user and group that own a file.

15. What is the purpose of /etc?

/etc stores important system configuration files.

16. What is the purpose of /home?

/home contains the personal directories and files of regular users.

17. What is commonly stored under /var?

/var stores changing data such as system logs and other variable files.

18. Why should rm be used carefully?

Because deleted files usually do not go to recycle bin and can be difficult to recover.

19. Why are Linux permissions important?

Permissions control who can read, change or execute files and help protect system data.

20. How can incorrect permissions create a security risk?

Incorrect permissions may allow unauthorized users to access, modify, or execute sensitive files, scripts, or data.

2. PRACTICAL LAB

2.1 Linux File Management & Permissions Lab SAFETY: run this ONLY inside your own Linux VM / authorized training environment / Jayvik Labs lab. Do NOT modify system files. Navigation + these file commands on your own lab files only.

No root privileges needed. SETUP: `cd ~ · mkdir day10_lab · cd day10_lab · pwd` (verify you are inside the lab directory)

```
Type 'help' to get help.

(kali㉿kali)-[/home/kali]
PS> cd ~

(kali㉿kali)-[/home/kali]
PS> cd ~

(kali㉿kali)-[/home/kali]
PS> mkdir day10_lab

(kali㉿kali)-[/home/kali]
PS> cd day10_lab

(kali㉿kali)-[/home/kali/day10_lab]
PS> pwd

Path
/home/kali/day10_lab
```

TASK 1 — Create files: touch notes.txt · touch report.txt · ls → you should see both files.

```
Path
/home/kali/day10_lab

(kali㉿kali)-[/home/kali/day10_lab]
PS> touch notes.txt

(kali㉿kali)-[/home/kali/day10_lab]
PS> touch report.txt

(kali㉿kali)-[/home/kali/day10_lab]
PS> ls
notes.txt  report.txt

(kali㉿kali)-[/home/kali/day10_lab]
PS> 
```

TASK 2 — Create directory: mkdir evidence · ls.

```
(kali㉿kali)-[/home/kali/day10_lab]
PS> mkdir evidence

(kali㉿kali)-[/home/kali/day10_lab]
PS> ls
evidence  notes.txt  report.txt
```

TASK 3 — Add content: echo "Jayvik Labs Day 10 Linux Lab" > notes.txt → cat notes.txt. The > writes command output into a file (lab files only).

```
(kali㉿kali)-[/home/kali/day10_lab]
PS> echo "Jayvik Labs Day 10 Linux Lab" > notes.txt

(kali㉿kali)-[/home/kali/day10_lab]
PS> cat note.txt
/usr/bin/cat: note.txt: No such file or directory

(kali㉿kali)-[/home/kali/day10_lab]
PS> cat notes.txt
Jayvik Labs Day 10 Linux Lab
```

TASK 4 — Copy: `cp notes.txt backup.txt` · `ls` → `backup.txt` exists.

```
(kali㉿kali)-[/home/kali/day10_lab]
PS> cp notes.txt backup.txt

(kali㉿kali)-[/home/kali/day10_lab]
PS> ls
backup.txt  evidence  notes.txt  report.txt
```

TASK 5 — Move: `mv report.txt evidence/` · `ls evidence` → `report.txt` moved in.

```
(kali㉿kali)-[/home/kali/day10_lab]
PS> mv report.txt evidence/

(kali㉿kali)-[/home/kali/day10_lab]
PS> ls evidence
report.txt
```

TASK 6 — Rename: `mv backup.txt notes-backup.txt` · `ls`.

```
(kali㉿kali)-[/home/kali/day10_lab]
PS> mv backup.txt notes-backup.txt

(kali㉿kali)-[/home/kali/day10_lab]
PS> ls
evidence  notes-backup.txt  notes.txt
```

TASK 7 — View permissions: `ls -l notes.txt` → identify Owner / Group / Others permissions.

```
(kali㉿kali)-[/home/kali/day10_lab]
PS> ls -l notes.txt
-rw-rw-r-- 1 kali kali 29 Sep 17 10:53 notes.txt
```

TASK 8 — Change permissions: `touch demo.sh` · `ls -l demo.sh` · `chmod u+x demo.sh` · `ls -l demo.sh` → what changed?

```

(kali㉿kali)-[/home/kali/day10_lab]
PS> touch demo.sh

(kali㉿kali)-[/home/kali/day10_lab]
PS> ls -l demo.sh
-rw-rw-r-- 1 kali kali 0 Sep 17 10:58 demo.sh

(kali㉿kali)-[/home/kali/day10_lab]
PS> chmod u+x demo.sh

(kali㉿kali)-[/home/kali/day10_lab]
PS> ls -l demo.sh
-rwxrw-r-- 1 kali kali 0 Sep 17 10:58 demo.sh

```

TASK 9 — Numeric permissions: `chmod 644 notes.txt` · `ls -l notes.txt` → explain Owner r/w, Group r, Others r.

```

(kali㉿kali)-[/home/kali/day10_lab]
PS> chmod 644 notes.txt

(kali㉿kali)-[/home/kali/day10_lab]
PS> ls -l notes.txt
-rw-r--r-- 1 kali kali 29 Sep 17 10:53 notes.txt

```

TASK 10 — Safe delete: `touch delete-me.txt` · `ls` · `rm delete-me.txt` · `ls`. LAB FILE ONLY — always verify the name before `rm`.

```

(kali㉿kali)-[/home/kali/day10_lab]
PS> touch delete-me.txt

(kali㉿kali)-[/home/kali/day10_lab]
PS> ls
delete-me.txt  demo.sh  evidence  notes-backup.txt  notes.txt

(kali㉿kali)-[/home/kali/day10_lab]
PS> rm delete-me.txt

(kali㉿kali)-[/home/kali/day10_lab]
PS> ls
demo.sh  evidence  notes-backup.txt  notes.txt

```